

Avi Vyas

✉ avi.vyas.work@gmail.com ☎ 9571790364 📍 Gurgaon, India 🌐 linkedin.com/in/avi-vyas

🔗 github.com/avi-vyas 🔗 avi-vyas.github.io/portfolio-application

PROFESSIONAL EXPERIENCE

Software Engineer (Backend Engineer)

01/2025 – Present | Noida, India

To The New (TTN) 🔗

- Developed and maintained scalable backend microservices using **Java** and **Spring Boot** for a platform serving **11.8M+ users**.
- Designed and optimized **Elasticsearch**-powered **autocomplete** and **global search** APIs by implementing fuzzy matching, multi-parameter queries, pagination, indexed sorting, and constant-score queries while migrating filtering, sorting, and business logic from Java to Elasticsearch, reducing average search latency by **over 90% (1.2s to <100ms)**.
- Developed an automated **free channel upgrade** solution that enabled eligible subscribers to upgrade to **HD channels** at the same subscription price across multiple touchpoints. Later generalized the functionality into a **reusable microservice** consumed by multiple teams, eliminating duplicate implementations.
- Architected and developed **configurable promotions** across **three implementation phases**, enabling recurring and multiple promotion types with **business-rule-driven configuration** across various customer touchpoints, reducing dependency on code changes for new promotional campaigns.
- Participated in **end-to-end feature development**, including requirement analysis, effort estimation, technical feasibility assessment, implementation, production deployments, and production issue resolution across multiple microservices.

SKILLS

Programming Languages — Java (primary), Python (FastAPI)

Frameworks & Libraries — Spring Ecosystem (Spring Boot, Spring AI, Spring Security, Spring Data JPA, Hibernate)

Databases & Messaging — MySQL, PostgreSQL, Redis, Elasticsearch, RabbitMQ

Cloud & DevOps — AWS (EC2, S3), Docker, Git, GitLab, Maven, Gradle

Testing & Observability — JUnit, Mockito, SLF4J Logging, Production Debugging

Core Concepts — Data Structures, OOP, SOLID, Microservices, Multithreading, REST API Design

PROJECTS

E-Commerce Backend System 🔗

Java, Spring Boot, MySQL, Spring Security, JWT

- Developed a **role-based backend system** (Admin, Seller, Customer) with secure authentication and authorization using **Spring Security** and **JWT**.
- Designed and implemented **RESTful APIs** for product management, user management, and catalog services.
- Modeled complex product structures with **metadata-driven design**, enabling flexible and scalable catalog management.
- Implemented **pagination, validation, and exception handling** for robust API design.
- Followed **layered architecture and clean code principles** for maintainability and scalability.

Doctor Appointment System 🔗

Python, FastAPI, MySQL, SQLAlchemy, JWT

- Built RESTful APIs for **appointment scheduling and booking systems** using FastAPI.
- Implemented **role-based access control (RBAC)** for doctors and patients with JWT authentication.
- Designed **time-slot scheduling system** with conflict detection and appointment lifecycle management.
- Used **ORM (SQLAlchemy)** for database interactions and modular architecture for maintainability.

EDUCATION

Bachelor of Technology (B.Tech) in Computer Science and Engineering

2021 – 2025

BML Munjal University

CGPA: 8.14 / 10